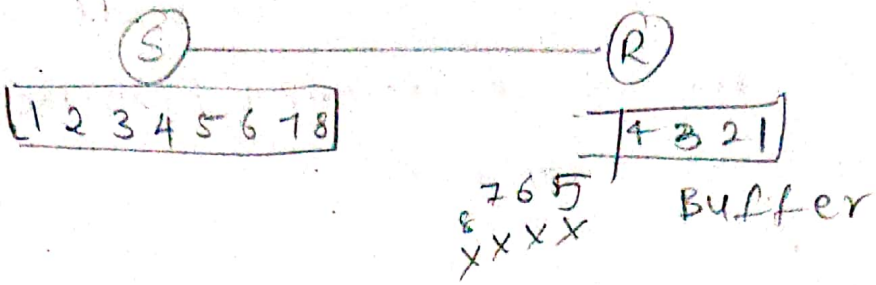
Flow control



→ Sender cannot sent the packets that a rate which receiver cannot handle them.

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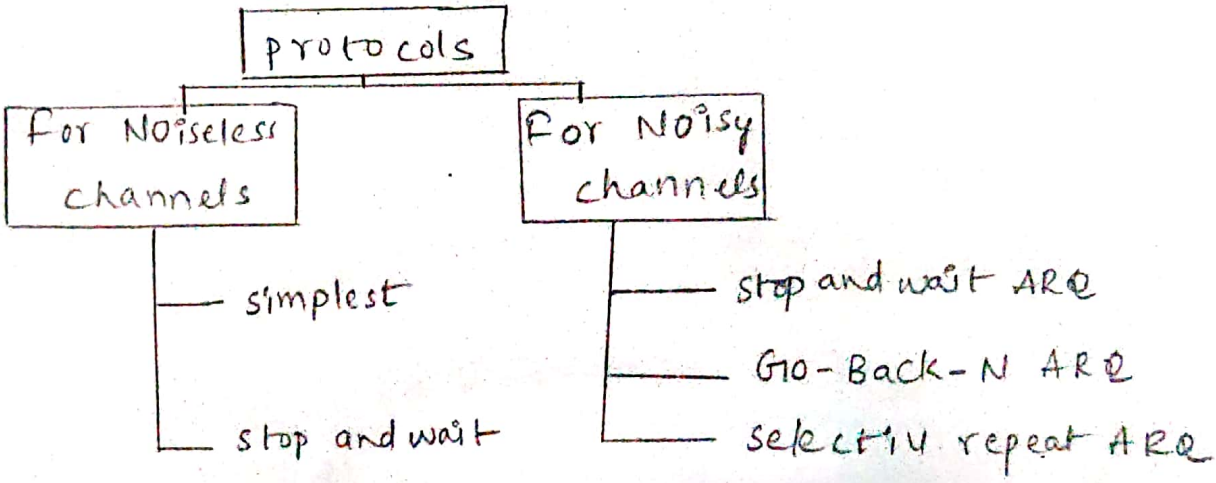
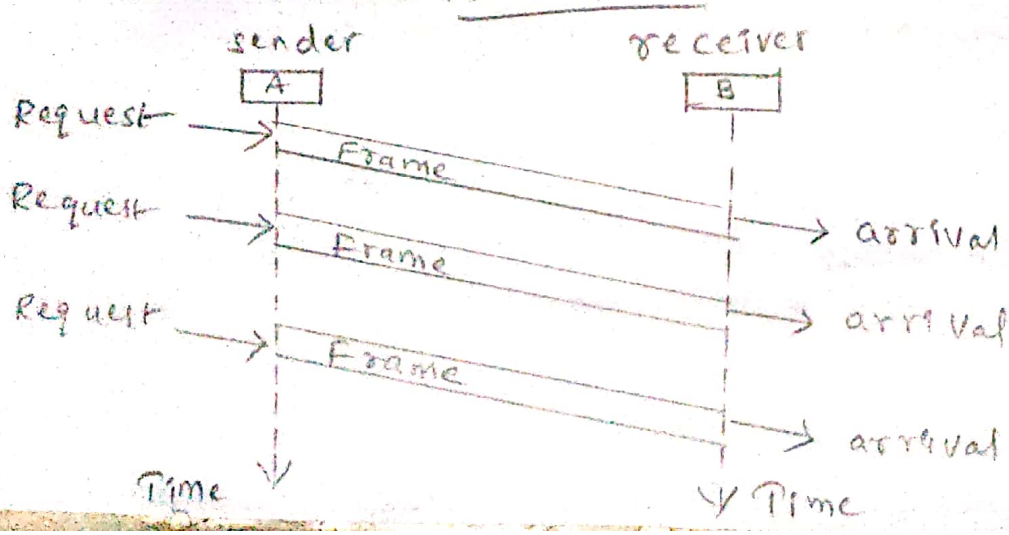


Simplest protocol :

In simplest protocol, the sender sends its frames one after another with no regards to the receiver. The ~~En stop o~~ simplest protocol has neither flow control nor error control. Flow diagram



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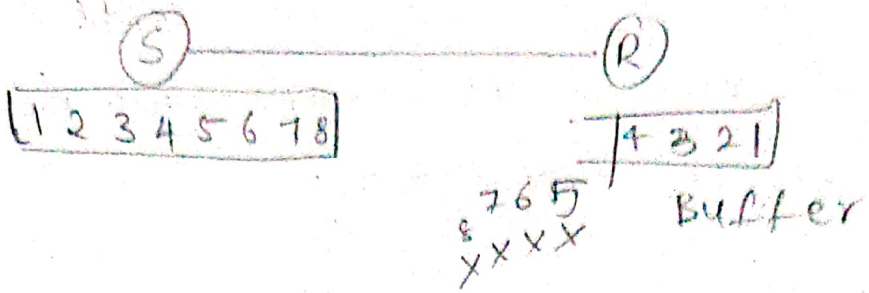
Flow control



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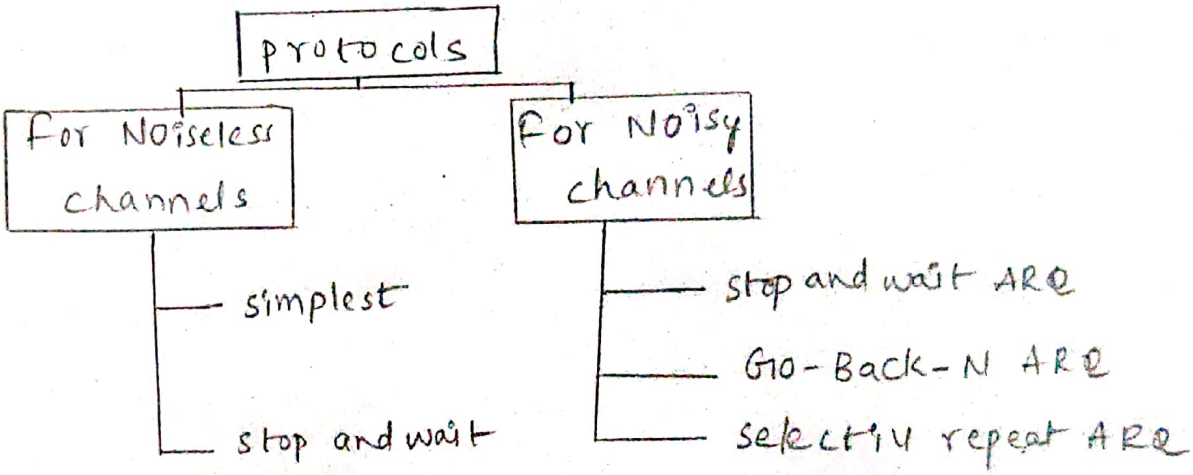

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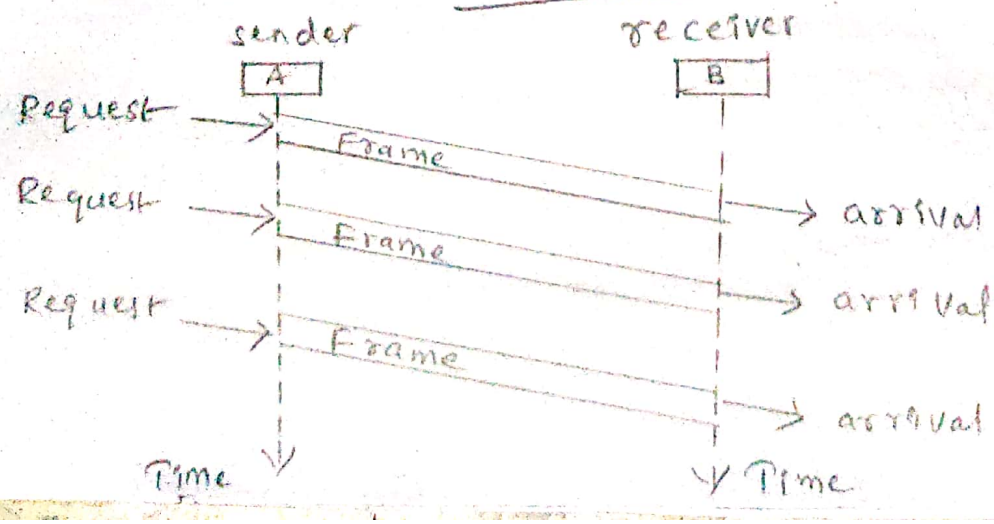

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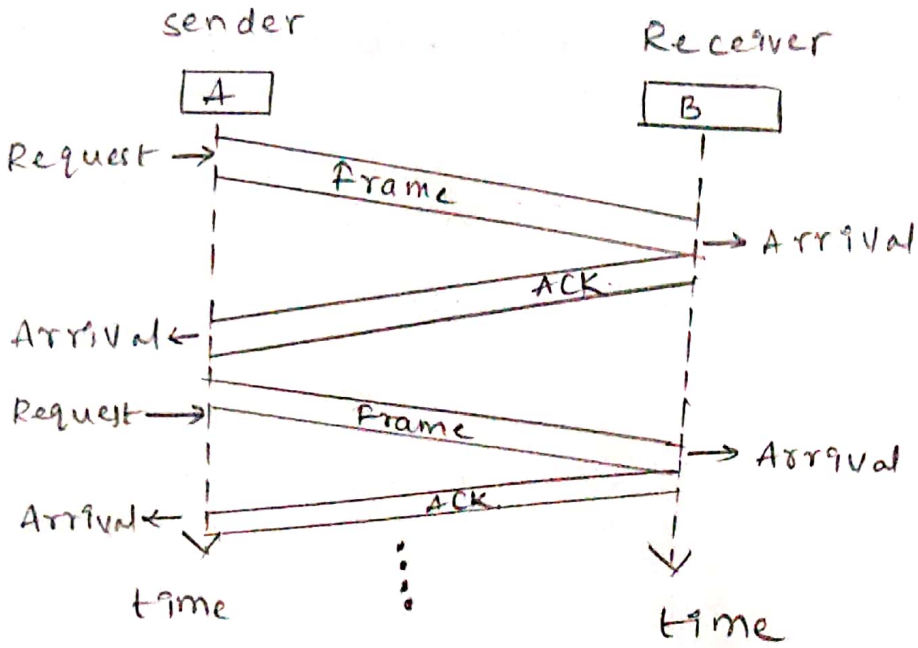
Stop and wait protocol :

In stop and wait protocol, the sender sends one fram stops until it receives confirmation from the receiver, and then sends the next frame.



flow diagram

→ The advantage of stop and wait is very simple to imple-ment.

→ Stop-and-wait protocol provides flow control.

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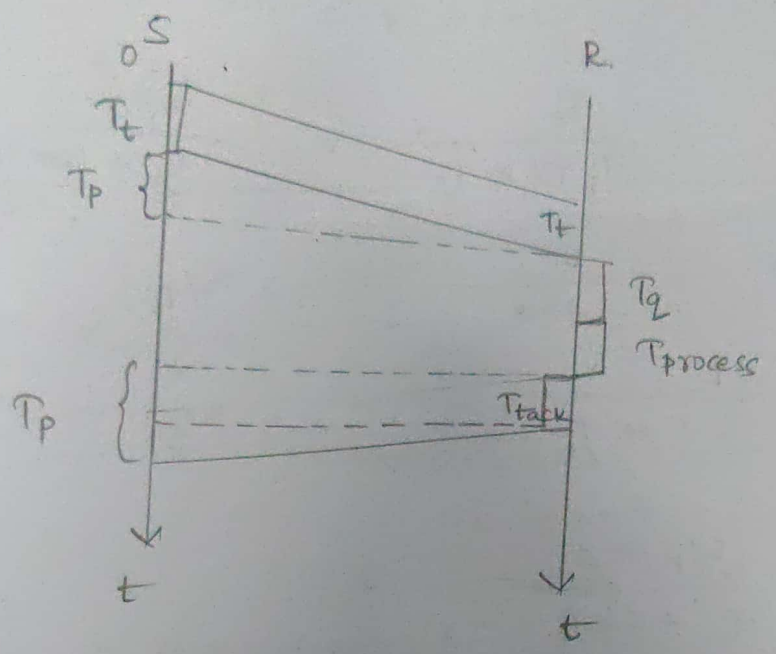

Total time (taken to send one packet) $= T_{tdata} + T_p + \cancel{T_q}^{\,0} + \cancel{T_{process}}^{\,0} + T_{tack} + T_p$

$= T_{tdata} + 2 * T_p + T_{ack}$

→ propagation delay will be same for data packet and ack.

→ In most of the cases the acknowledgement size is very small compare to the data size, so we ~~neg~~ ignore transmission delay for acknowledgement. In piggy backing we will attach ack on the data packet. In that can the ack transmission time will be equal to transmission time of data packet.

$$\boxed{\text{Total time} = T_t + 2T_p}$$

→ In stop and wait we are actually T_t time efficiently and ~~2x~~ we ~~are~~ waiting / waiting $2 * T_p$ time.

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$$\text{efficiency } (\eta) = \frac{\text{useful time}}{\text{total cycle time}}$$

$$= \frac{T_t}{T_t + 2 * T_p} = \frac{T_t / T_t}{\frac{T_t}{T_t} + \frac{(2 * T_p)}{T_t}}$$

$$\eta = \frac{1}{1+2a} \quad (\because a = T_p / T_t)$$

effective Bandwidth (or) Throughput $= \dfrac{L}{T_t + 2 * T_p}$

Bandwidth utilization

$$= \frac{(L/B) * B}{T_t + 2 * T_p}$$

$$= \frac{T_t}{T_t + 2 * T_p} * B$$

$$= \left(\frac{1}{1+2a}\right) B$$

$$\boxed{\text{Throughput} = \eta * B}$$

→ If the transmission time is more than 2* propagation time then the efficiency will be more.

$$T_t \geq 2 * T_p \rightarrow \text{more efficiency}$$

$$\frac{L}{B} \geq 2 * T_p \quad \left(\because T_t = \frac{L}{B}\right)$$

$$\boxed{L \geq 2 * T_p * B}$$

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The efficiency of stop-and-wait depends on few factors

$$\eta = \frac{1}{1+2a}$$

$$\eta = \frac{1}{1+2*\frac{T_p}{T_t}} = \frac{1}{1+2*\frac{d}{v}*\frac{B}{L}}$$

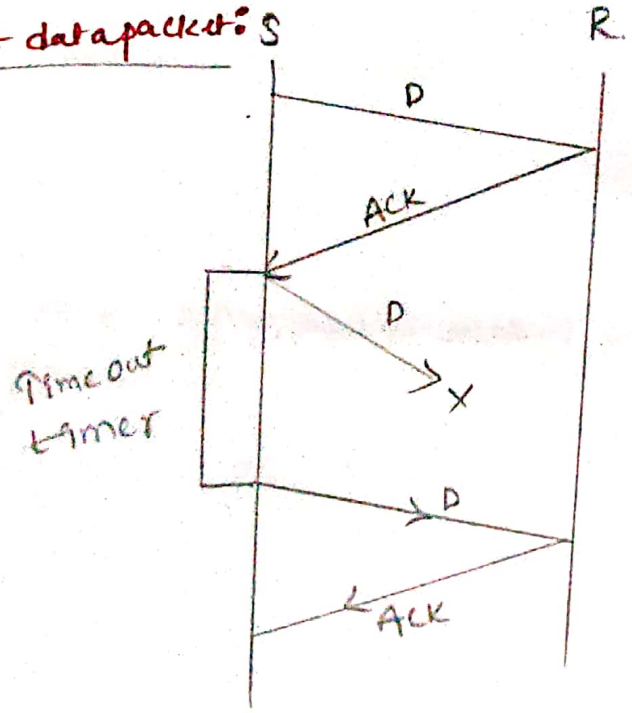
i) $d \uparrow \Rightarrow \eta \downarrow$

ii) $L \uparrow \Rightarrow \eta \uparrow$

Hence stop-and-wait is more suitable than LANS compare to WANS.
If the packet length is more ~~busy~~ then the efficiency will be more.

problems in stop and wait :-

1) Lost data packet:



→ if we add time out timer to stop and wait then it becomes stop-and-wait ARQ. [Automatic repeat request]

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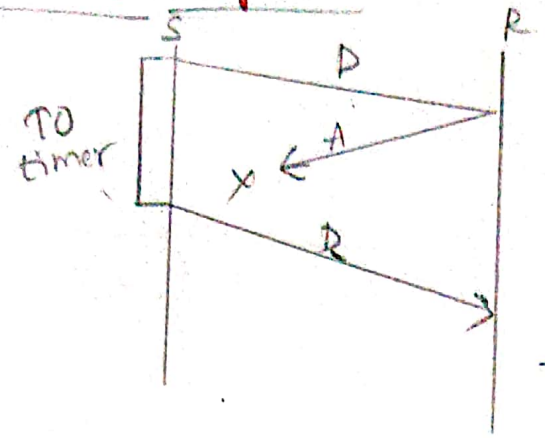
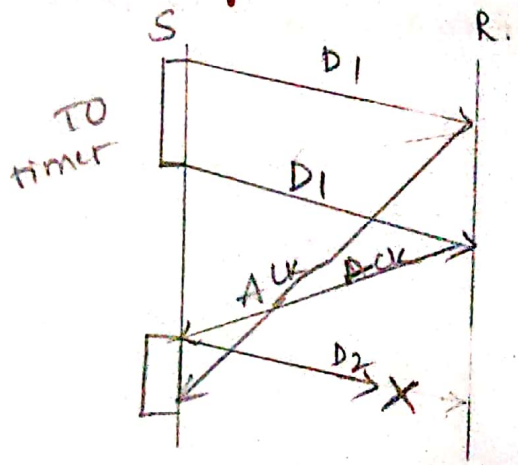
Lost Acknowledgement :-



→ Duplication of data at receiver.

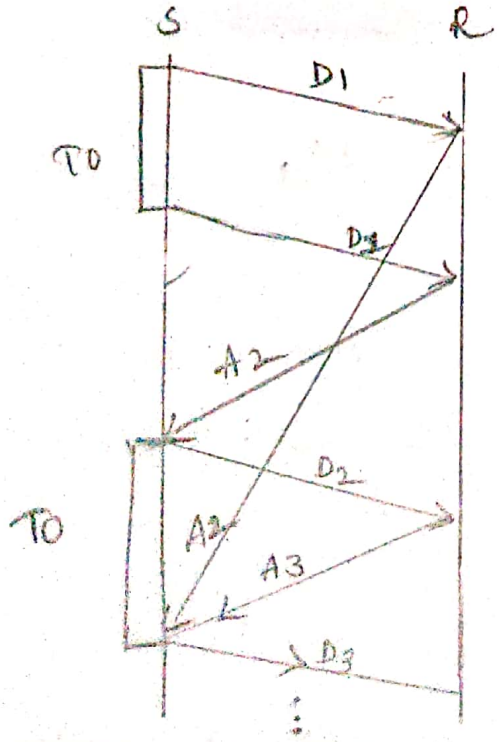
→ S&W + TO + sequence no's. to data.

Delayed Acknowledgement :



⇒ Missing data problem.

→ S&W + TO + sequence no. to data + sequence no. on acknowledgement.



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...ining :

Capacity of a channel = Bandwidth * propagation time (Half duplex)

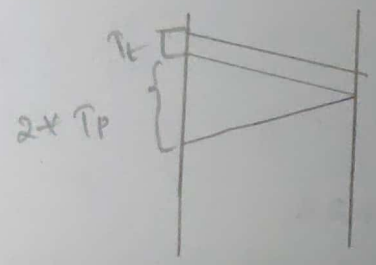
= 2* bandwidth * propagation time (full duplex).

→ If channel capacity is very high they are called as thick pipes.

→ If the channel capacity is low they are called as thin pipes.

→ Stop-wait is suitable for thin pipes.

$$\eta = \frac{1}{1+2a} = \frac{1}{1+\frac{2*T_p*B}{L}}$$ → capacity of channel.



if T_t = 1 packet

$2*T_p$ – ?

T_t sec – 1 packet

1 sec – $\frac{1}{T_t}$ packets

$(T_t + 2*T_p)$ – $\frac{T_t + 2*T_p}{T_t}$ packets

– $(1+2a)$ packets.

Actually whatever the time we are ~~sending~~ spending to send one packet, in that time we can send $(1+2a)$ packets.

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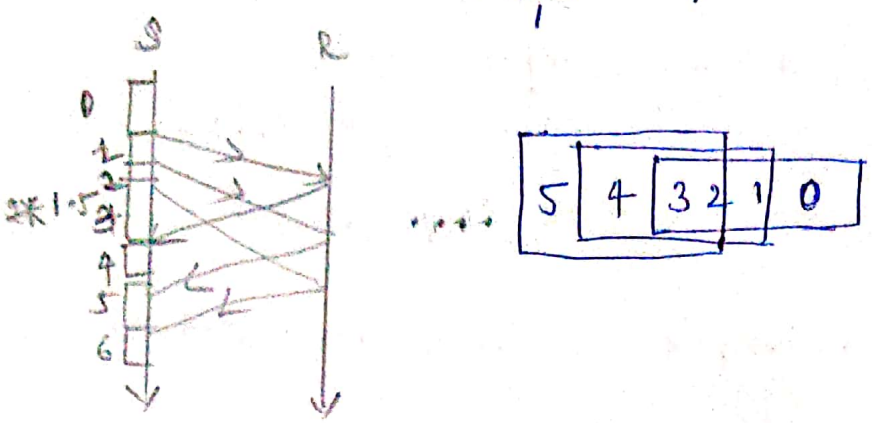
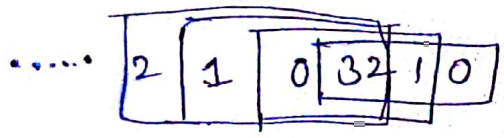
ex: $T_t = 1ms$, $T_p = 1.5ms$

$$\eta = \frac{1}{1+2a} = \frac{1}{1+2\times\frac{1.5}{1}} = \frac{1}{4}$$



→ The sender window size should $(1+2a)$ for maximum efficiency.

→ We can reuse the sequence numbers.



$W_s = 1+2a$

min sequence no's required $= 1+2a$

bit seq number field $= \lceil \log_2(1+2a) \rceil$

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back N:

Sliding window protocol

- Go Back N ARQ
- selective repeat ARQ

→ sender window size in GBN is N. sender window size should be always greater than 1.

if $T_t = 1ms$, $T_p = 49.5ms$, $\eta = ?$, GBN, $B = 40mbps$

$W = 1 + 2a = 100$

$N = 10$

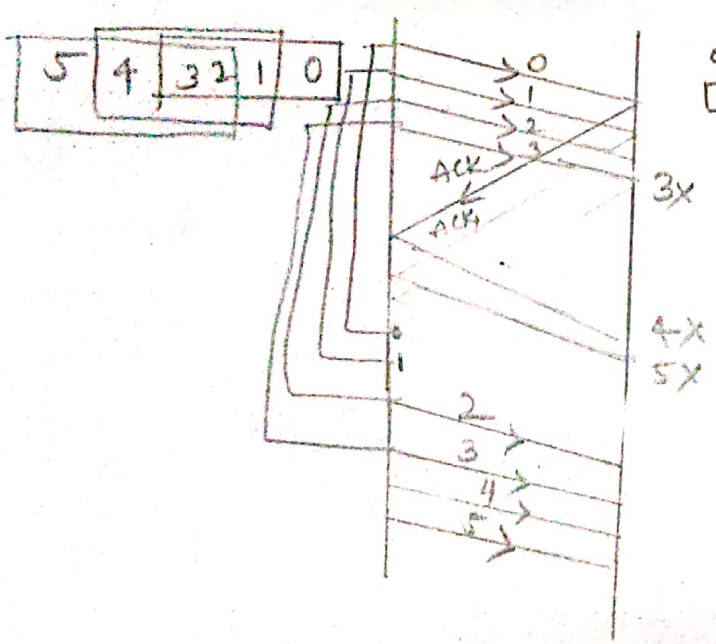
$\eta = \frac{10}{100} = 10\%$,

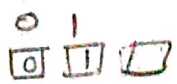
Throughput $= BW \times \eta$

$= \frac{10}{100} \times 40$

$= 4$

→ Receiver window size is always 1.





2 he didnot receive so he will discard 3, 4, 5

2, 3, 4, 5 has to be retransmit according to sliding window.

from 5 we should go back (last packet in the window).

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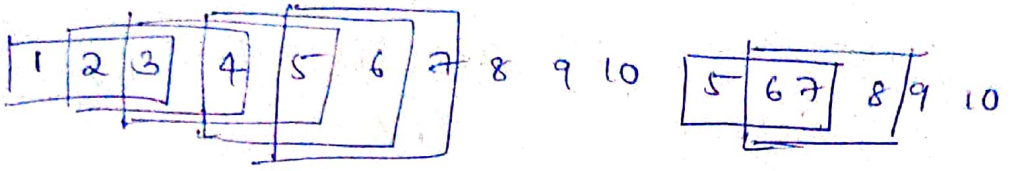
In GB3, if every 5th packet that is being trans is lost and if we have to send 10 packets, then how ma transmissions are required.



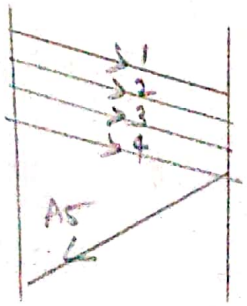
5 lost

retransmission from last packet.

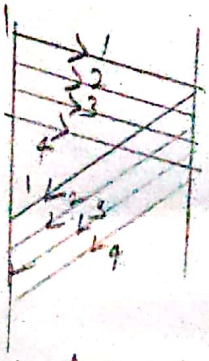
(18)

→ In GBN receiver will never accept out of order packets.

3)



cummulative ACK



independent ACK

adv : traffic is less

disadv : less reliability

(if ack lost it means all packets lost)

disadv) traffic is high

adv) reliability is more.

→ In GoBack N we will send one acknowledgement for many packets.

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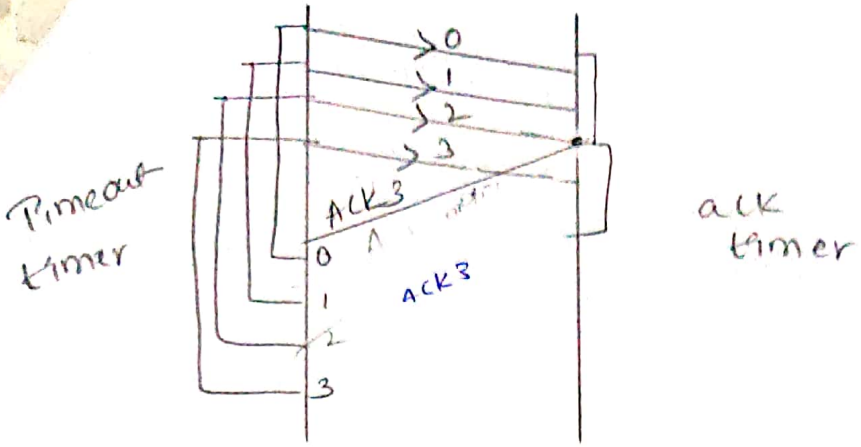
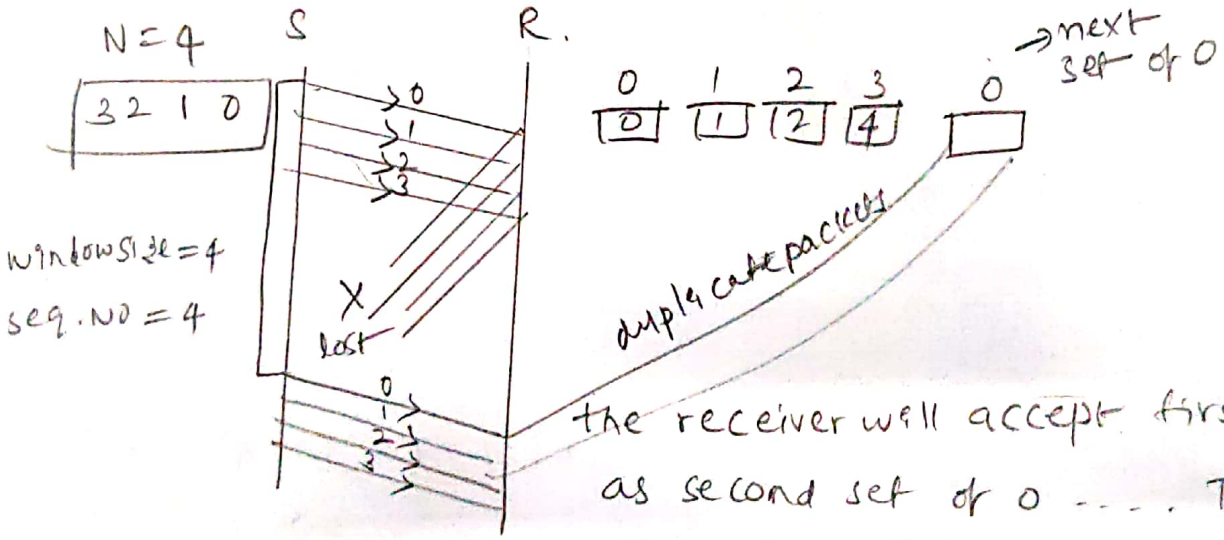


→ at the receiver side the receiver maintains ack timer, within ack timer whatever the packets received by receiver to all those packets the receiver sends one ACK.

To timer > ACK timer.



the receiver will accept first set of 0 as second set of 0 .... The packet will be duplicated at receiver side

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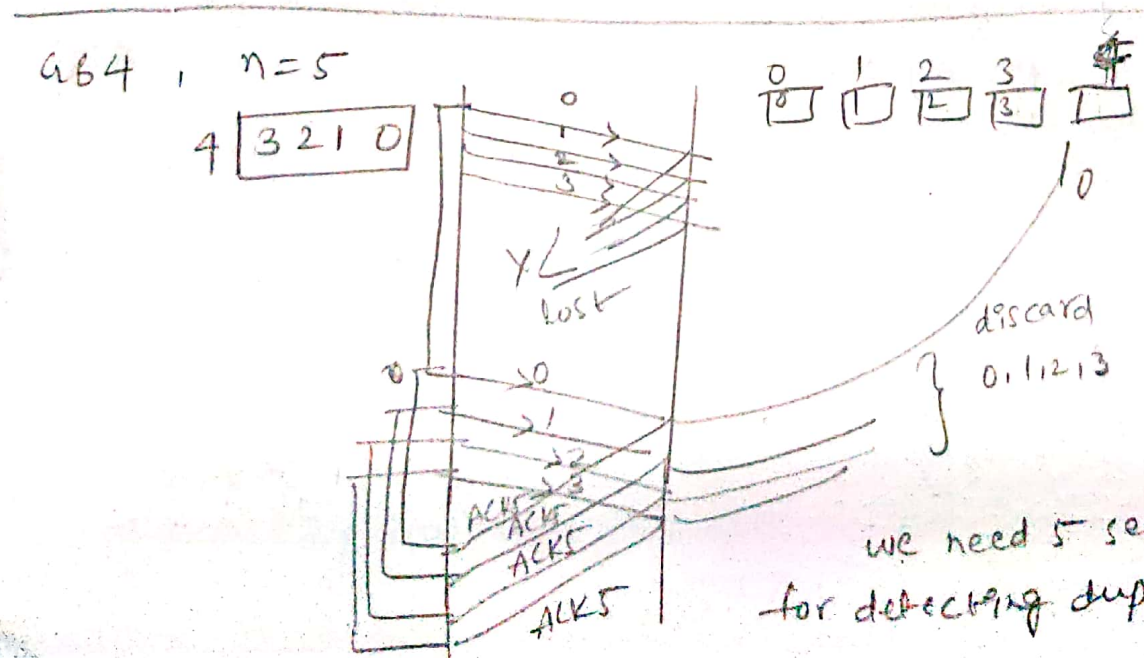

we need 5 sequence nos. for detecting duplicate packets.

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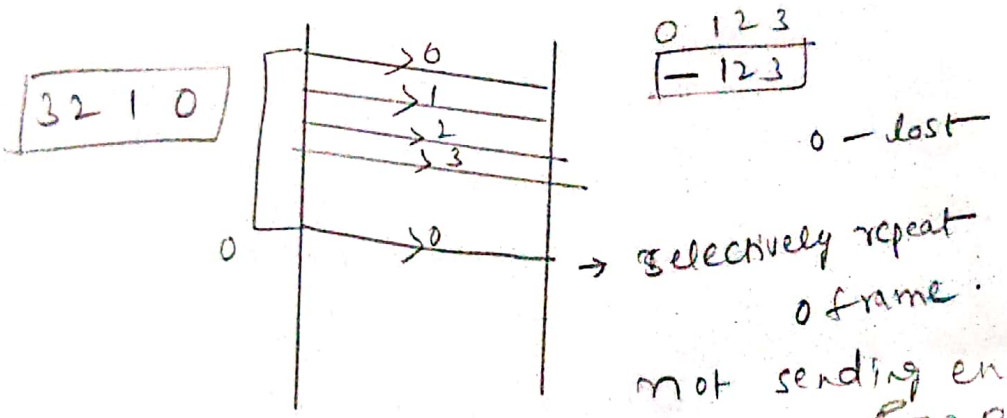
→ if the sender window size N, Receiver window size is 1 then for detecting duplicate packets we need (N+1) sequence numbers.

$$(W_S + W_R) \leq ASN \quad \text{(available sequence no's)}$$

SR

1) sender $W_S > 1$
2) receiver window size is $W_R = W_S$
3)



→ selectively repeat 0 frame.

not sending entire window like Go Back N.

→ Retransmissions are very less in SR ARQ compare to Go Back N.

→ when the packet is corrupted/lost in SR, the receiver will send a NAK by that the sender sends the packet again.

→ ACK timers are not required in SR.

SR ⇒ sender → searching, Receiver → sorting